necessarily battery scientists, but I would guess that the answer is at least "some," if it turns out that it is possible to boost several consecutive cycles without compromising life.

In the end, I think that this paper can be viewed as an excellent example of how a simple idea, requiring no new technology, can break conventional paradigms to produce a cell that, at least plausibly, can be commercialized.

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Toward Sustainable H_2 Production: Linking Hydrogenase with Photosynthesis

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Molecular hydrogen, H_2, is an energy carrier that is increasing in popularity as an alternative fuel. Presently, the major part of commercially available H_2 is produced from fossil resources. To make H_2 a true contender to fossil fuels, a sustainable production via water splitting by renewable electricity in combination with earth-abundant catalysts or by photosynthetic microorganisms is required.

Biological H_2 production usually involves enzymes known as hydrogenases that are found in microorganisms. The majority of hydrogenases can be classified into two groups based on their active site: the NiFe hydrogenases that have a heterometallic active site with one Fe and one Ni ion and the FeFe hydrogenases, carrying an inorganic Fe_2 cofactor in the active site. Interestingly, each species of microorganism has either one or the other type, but never both.

In existing biogas plants, biomass—usually waste—is fermented by microorganisms to methane, CO_2 and H_2. Since biomass is produced by photosynthesis, this process indirectly utilizes solar energy. Many photosynthetic microorganisms carry hydrogenases and thus have the potential to produce H_2 directly from solar energy in a cleaner, more efficient way than by fermentation. This was described first by Gaffron, who discovered H_2 production by green algae in 1937.[1] Ever since, it has been a dream to create a direct link between photosynthesis and hydrogenases to drive solar H_2 production from water (reaction 1 in Scheme 1 and Figure 1).

Oxygenic photosynthesis is powered by light-driven charge separations in two reaction centers (Figure 1). Photosystem II (PSII) oxidizes water and reduces the plastoquinone (PQ) pool while releasing molecular oxygen and protons (reaction 2). The electrons are delivered to photosystem I (PSI) via

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$$2\,H_2O + 8h\nu \rightarrow O_2 + 2\,H_2 \qquad (1) \qquad E^{\circ} = +1.23\ V$$

$$2\,H_2O + 4\,h\nu \rightarrow O_2 + 4\,H^+ + 4\,e^- \qquad (2) \qquad \text{PSII, } E^{\circ\prime} = +\,820\ mV$$

$$4\,e^-\ (0\ mV) + 4\,h\nu \rightarrow 4\,e^-\ (-\,430\ mV) \qquad (3) \qquad \text{Plastoquinol (PSII)} \rightarrow \text{Fd (PSI)}$$

$$4\,H^+ + 4\,e^- \rightarrow 2\,H_2 \qquad (4) \qquad \text{Hydrogenase, } E^{\circ\prime} = -\,410\ mV$$

Scheme 1. Reactions Involved in Photosynthetic Water Oxidation and Hydrogen Production
The overall reaction (1) occurs naturally in photosynthetic organisms only via a number of sequential photochemical and metabolic reactions. The redox potentials for the biological components are stated for pH 7, indicated by the "prime" sign.

the cytochrome (Cyt) b_6f complex and plastocyanine (Pc). In PSI, the second light reaction results in a highly reducing electronic state, which leads to reduction of ferredoxin (Fd, reaction 3). As seen from Scheme 1, only after this second light reaction will there be electron carriers that are reducing enough to drive CO_2 reduction via the Calvin-Benson-Bassham (CBB) cycle or, alternatively, H_2 production by hydrogenases (reaction 4).

Because ferredoxin can function as electron donor to most hydrogenases, there is a direct, natural path for electrons extracted from water by light-driven charge separations in PSII to H_2 production by hydrogenases. However, there are two main obstacles for realizing a high efficiency for this "dream" reaction: (1) the most active hydrogenases are highly oxygen sensitive, since O_2 leads to "poisoning" of their active site, and (2) the competition for highly reducing electrons between carbon fixation by the CBB cycle and H_2 production (Figure 1). For photosynthetic microorganisms, CO_2 fixation is required for growth and thus is the main target for electrons extracted from water. Only when this metabolic "highway" becomes congested may H_2 production act as an elegant valve to relieve the electron pressure and avoid stress.

Initial approaches to circumvent these obstacles involved a temporal separation of photosynthetic growth and H_2 production. PSII activity, and thus the presence of O_2, was suppressed after a photosynthetic growth phase by, for example, sulfur deprivation.[2] Under such induced anaerobic conditions, starch degradation serves as one of the electron sources for H_2 production (Figure 1). From the point of view of energy conversion efficiency, this approach is still suboptimal, as it partially involves the intermediate storage of solar energy in biomass (starch) and because the H_2 production is not continuous.

Thus, even more direct approaches are desirable. Several attempts have been made to achieve solar-powered H_2 production by introducing foreign or even semi-artificial hydrogenases into cyanobacterial cells[3] or by eliminating competing pathways. In an extension of the latter strategy, the direct coupling or tethering of the hydrogenase enzyme to the PSI reaction center is an attractive option. The idea is simple: the competition of hydrogenase and the CBB cycle for ferredoxin is eliminated once the relatively small hydrogenase is attached to PSI close to the exit point of electrons from the reaction center. From a thermodynamic point of view, this should work like a charm (Scheme 1). The difficulty with the approach instead lies in the complexity of biological systems. Previous attempts have therefore used isolated proteins that have been connected and used for light-induced H_2 production *in vitro*.[4]

Now a big leap has been taken by performing this linkage *in vivo* in both cyanobacteria and green algae.[5,6] Two separate studies, Kanygin et al., and Appel et al., show that it is possible to utilize the microorganisms own genetic material to "trick" the cells into making these chimeric proteins from scratch. Appel et al. utilized a cyanobacterium, *Synechocystis* PCC 6803. Cyanobacteria are the bacterial ancestors to plants

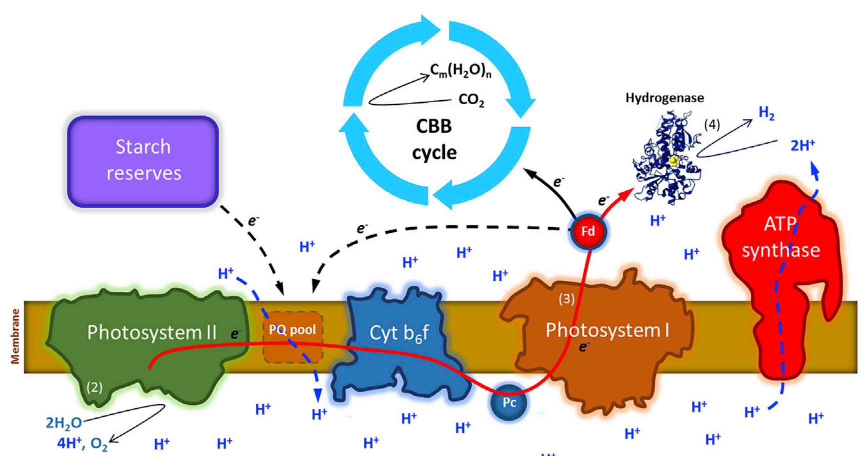

Figure 1. Simplified Electron and Proton Transfer Pathways in Green Unicellular Algae and Cyanobacteria Leading to H_2 Production
Numbers represent reactions described in Scheme 1. The red arrow signifies the PSII and PSI driven multistep electron transfer from water to H_2 (reaction 1).

and are relatively easy to modify genetically. Cyanobacteria carry the less active and relatively large NiFe hydrogenases. Kanygin et al. instead used *Chlamydomonas reinhardtii*, which is a eukaryotic green alga. This means that it has a more complex cell structure and therefore is significantly more difficult to make genetic alterations to. On the upside, *Chlamydomonas* carries an FeFe hydrogenase, which is a smaller protein, and thereby the distance for electron transfer from PSI to the nearest 4Fe4S cluster in hydrogenase can be kept short (15 Å versus 23–45 Å in Synechocystis), ensuring a more efficient electron transfer.

In both cases, the authors made use of the fact that PSI comes with several small protein subunits on the reductive (''acceptor'') side. These are made from their own genes, implying that they can be altered or replaced without changing the main body of PSI. To make the alterations possible, each of the studies involved removing the gene encoding one of the PSI subunits and replacing it with a gene that codes for the very same protein, but now with a bonus attachment in the form of a hydrogenase.

Both studies demonstrate direct light-driven electron transfer from water oxidation in PSII to hydrogenase, in addition to H_2 production due to other metabolic processes. After a dark period, photoH_2 production occurs in two waves: a more intense first transient (max 1 h) followed by a slower, long-lasting production (6–20 h). The maximum rates of this latter, arguably more relevant phase were 25 μmol H_2 h^{-1} mg(Chl)$^{-1}$ (Kanygin et al.)[5] and 0.63 μmol H_2 h^{-1} mg(Chl)$^{-1}$ (Appel et al.).[6] It is noted that attaching hydrogenase to PSI does not remove the second bottleneck of photoH_2 production, the oxygen sensitivity of the hydrogenases in these systems. Thus, Appel et al. employ an enzymatic oxygen scavenging assay to continuously remove O_2 from the medium. By contrast, this was not necessary in the study by Kanygin et al., since the PSI-hydrogenase construct was expressed at 7-fold lower yields than PSI in the wild-type cells, which strongly reduced the ratio of PSI to PSII. As shown previously, this imbalance results in a reduced PQ pool and in turn in a diminished O_2 production. Thus, the O_2 produced by PSII is largely consumed internally by respiration.[7] While the cyanobacterial system was able to grow photoautotrophically, the green algae was grown on acetate, which may have additionally reduced PSII activity.

It is useful to compare the above H_2 production rates with other recent alternative approaches for direct solar H_2. These include restricting linear electron flow by (1) changing the PSII/PSI ratio,[7] (2) combining CO_2 limitation and use of oxygen absorbents,[8] and (3) using short light/dark intervals that allow PSII activity without onset of CO_2 fixation.[9] All these treatments resulted in the reduced state of the PQ pool, which promotes hydrogenase activation.[10] Some of these approaches resulted in H_2 production that was sustained over several weeks,[7] while others have comparable rates to the Kanygin et al. study.[8,9]

While there is still some way to go, these recent advances in linking hydrogenases to natural photosynthetic reactions brings optimism to the development of direct biological photoH_2 production. Improving reaction center fusion constructs, as well as their combination with the various alternative approaches described above, may lead to a further improvement of the efficiency and sustainability of this process. However, reducing or even eliminating the oxygen sensitivity of hydrogenases while retaining high activity may be required to pave the way toward future commercial exploitation of photobiological H_2 production.

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